

Week 1 — Introduction to Business Forecasting

Business Forecasting

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Who am I?

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A decision you'll be able to make

You manage inventory for Walmart México. It's April. How many umbrellas should each store stock for June?

- Order too few → empty shelves, lost sales, unhappy customers.
- Order too many → cash tied up in stock that sits until next year.

The right number is a **forecast**: last year's June sales, the trend, the rainy-season pattern, a promotion you're planning — turned into a decision, with a sense of how uncertain it is.

By the end of this course, you'll build exactly this kind of model — on your own data.

Where this course is going

Every week is one step from *raw data* to a *forecast you can defend*:

Describing data → Reasoning under uncertainty → Regression → Time series forecasting

Week 1 of describing data. Today: what forecasting is, what it is used for, and how this course works. From next week we work with real data in R.

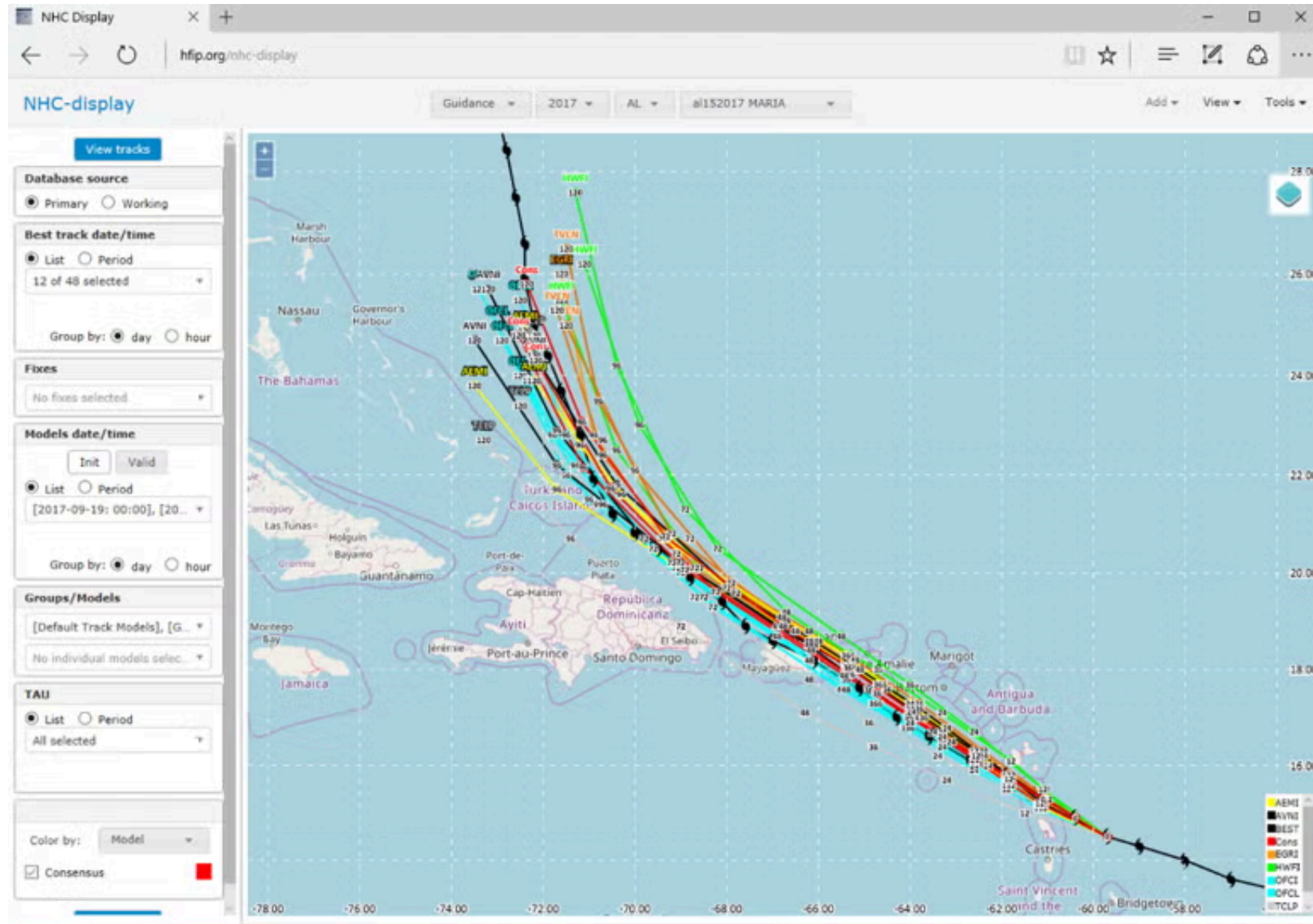
What Is This Class About?

Forecasting is the science and art of making predictions about future events using historical data and relevant information.

In this class, you will learn how to turn data into actionable insights that drive better decisions in business and beyond.

Why is forecasting important? Weather

Predicting hurricanes → timely reactions and preparations.

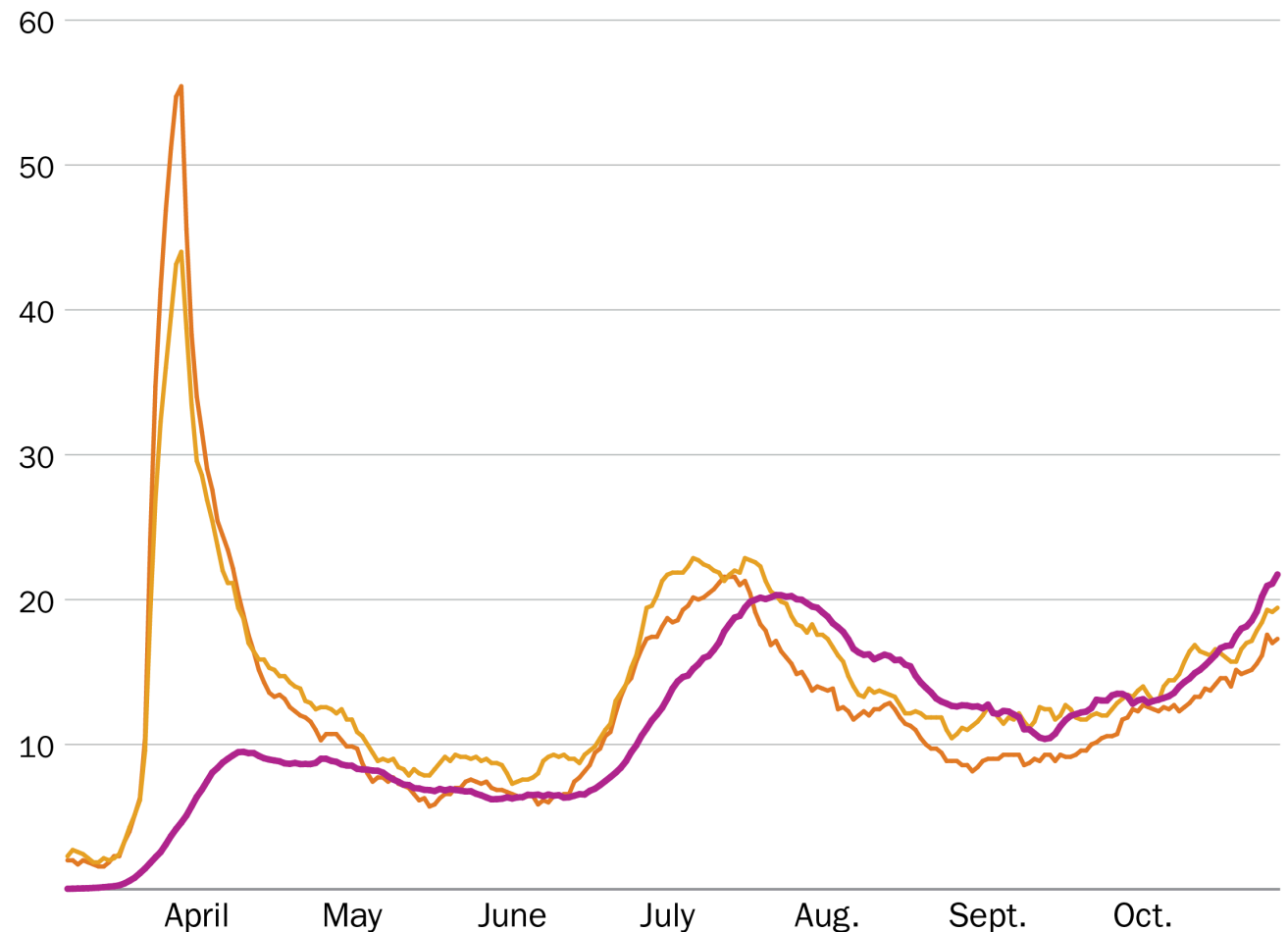


Epidemic forecasting

Predicting disease outbreaks → effective public health responses.

Google searches vs. cases: United States

— Seven-day average of new cases per 100,000 residents
Seven-day average of indexed searches for: — “loss of smell” — “loss of taste”

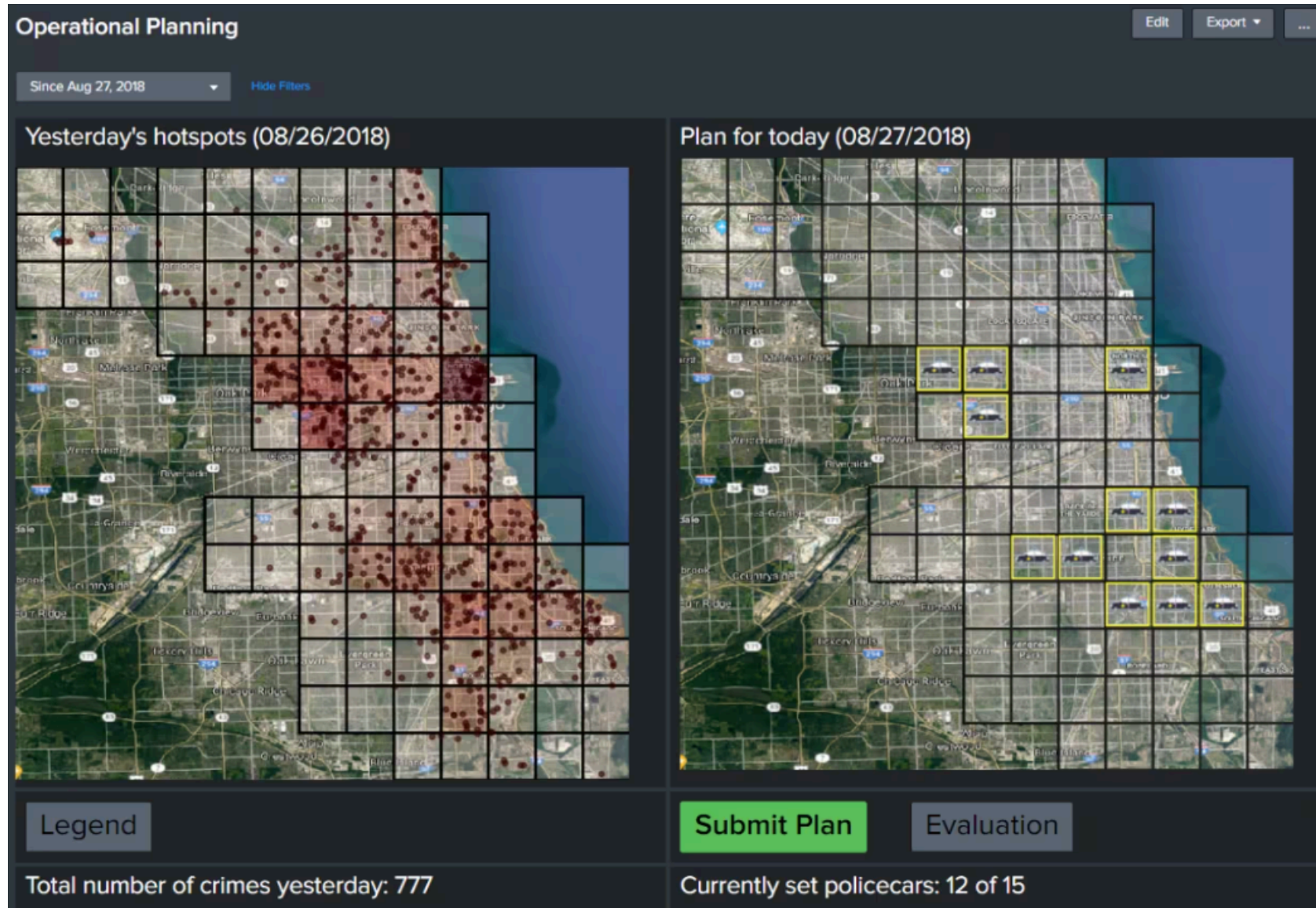


Sources: Google Trends, Post data

THE WASHINGTON POST

Crime forecasting

Identifying crime hot spots → more effective police patrols, and individuals at risk → targeted interventions.



Predicting 911 calls in Mexico City

One of your tasks later in class: **Predict 911 calls in Mexico City**



Forecasting in Business

- With the increasing availability of data, firms rely on rigorous data-driven methods to make decisions.
- Many business decisions depend on anticipating future events.
- In this course, you'll learn how to use data to make informed predictions about the future.

Applications of Forecasting

1. **Anticipating Demand** Adjust inventory to minimize waste and meet customer needs effectively *Example: Meal kits, retail chains*
2. **Employee Churn** Predict which employees are likely to leave and develop strategies to retain talent *Example: HR analytics in consulting firms*
3. **Strategic Investment Decisions** Use market forecasts to guide investment timing and location *Example: Expansion of electric vehicle infrastructure*
4. **Improving Advertisement Effectiveness** Identify which components of an ad drive conversions *Example: A/B testing by Airbnb*
5. **Product Recommendations** Forecast customer preferences based on past behavior *Example: Amazon, Netflix recommendation systems*

Your turn

- Get in pairs
- Consider your past employment or your future employment
- Think about how forecasting could solve some problems in the context of industry you are considering
- (5 min)

What will you learn?

1. Getting Business Information from the Data

- Analyze data, evidence, and arguments to make reasoned judgments

2. Problem Solving and Forecasting

- Formulate, evaluate, and implement statistical models for business forecasting.
- Interpret the results and validate assumptions
- Key technical skill very valuable on the job market!

3. Decision Making and Communication

- Choose optimal options to achieve objectives.
- Communicate findings, conclusions, and recommendations effectively to business professionals

Organization

- **Lectures Schedule:**
 - Monday and Wednesday Morning
- **Textbooks**
 - See canvas course materials and syllabus
- **Organization**
 - see the website

Grading

- **70% Exams**
 - 3 exams (2 midterms + cumulative final), applied, in the computer lab
- **20% Group Project** (one project, all semester)
 - A semester-long project with an **oral defense** — you explain your own work
- **10% Quizzes & Datatons**
 - 3 short in-class Canvas quizzes + 2 live “Datatons” (forecasting challenges)
 - **Quiz 1 (week 3) is an easy R/RStudio check — see below; lowest score dropped**

Everything that counts is done in class or defended in person.

Project checkpoints — put these in your calendar now

The project runs all semester, with checkpoints so no team discovers in November that its data doesn't exist:

Checkpoint	Date
Group formation (register your team on Canvas)	Tue 1 September
Topic & idea + dataset proposal (1 page)	Tue 29 September
Data-readiness check	Tue 27 October

Submission and the defenses happen in the last week of classes.

Getting RStudio working — this week, not in November

You are examined in RStudio, so the setup has to be boring by then. Two steps:

1. **RStudio_practice_1.R** — posted on Canvas now, **ungraded**. Open it in RStudio and run it line by line: load a file, count categories, take a mean and a median, draw two plots. Every command is on the cheat sheet.
2. **Quiz 1, week 3** — the same commands, in the lab, on your laptop, 20 minutes, cheat sheet allowed, **and you can ask me coding questions during it**.

It is deliberately easy. Its only job is to make sure that “R doesn’t work on my laptop” happens in week 3, when it costs nothing, instead of in the midterm.

Pre-requisites

- Mathematics III or Linear Algebra I
- Statistics II or Statistical Inference

Language

- Class is in English
- But your English is not evaluated
- If you don't know a word, feel free to ask
- Great occasion to learn vocabulary useful for interviews

AI Tools

- Most of your grade — **the exams and the project defense** — is earned **without AI**. You won't have access to it there, so you need to genuinely **learn and understand** how to do these things on your own.
- You **can** use AI to **support your learning**: refining code, summarizing concepts, checking your understanding.
- Be honest with yourself, though — students who leaned heavily on AI **without challenging themselves** typically did **not** do well in this course.

Use AI to learn, not to replace thinking

- Great for review — try asking:
 - “Why can’t we say a 95% confidence interval contains the true value with 95% probability?”
 - “What’s the relationship between Type I and Type II errors?”
 - “What is a p-value and what is it not?”

AI and learning

Explain p- value to me as Taylor Swift



Alright, let's do this the Taylor Swift way!

🎵 Hey, it's me, Taylor Swift! Let's talk about p-values like we're writing a song together. 🎵

So, imagine you're writing a song about someone who you *think* might be into you. But you're not sure, so you're looking for signs—like how often they text you, or if they show up at your shows. Your null hypothesis (just a fancy way of saying your starting assumption) is, "They're just being nice; they don't really like me."

Now, let's say you collect a bunch of "data" (you know, how often they text, how long they stay at your shows, whether they're in the front row singing every lyric). You run the numbers, and out comes the p-value.

If the p-value is super low (like 0.01), that's like saying, "There's only a 1% chance that all these signs are just a coincidence." In Taylor Swift terms, this is the part in the song where you're like, "*Oh my God, it's fate! They totally like me!*" So you reject the null hypothesis and start planning your next big love song.

But if the p-value is high (like 0.5), that's more like saying, "There's a 50/50 chance they're just being nice and it's nothing special." In this case, you might be singing something more like "*We are never ever getting back together*" because there's just not enough evidence to believe they're into you.

In short, the p-value helps you figure out whether to go all-in on those feelings or if you should just shake it off!

Academic Integrity

- Integrity matters — for your own learning, for your peers, and for the course.
- **We will employ a set of tools to detect cheating. Each episode of cheating will be reported.**

Software: R

Performing Real Forecasts with Real Data

- **Widely Used** R is one of the most popular languages for data analysis, statistical modeling, and forecasting — especially in academia and industry.
- **Free and Open-Source** R is completely free to use and supported by a large global community.
- **Great Community Support** Thousands of tutorials, packages, and forums are available — it's easy to find help when you're stuck.
- **Powerful Data Visualization** R includes powerful tools (like `ggplot2`) to create clear, publication-quality visualizations that help you communicate your insights effectively.

R is a tool, not the subject

This course is **not about learning R**. It is about forecasting — R is just the tool we use to do it.

So you are never asked to memorise code:

- The **R code and formula cheat sheets** are given to you — in class, in the practice sets, and **in the exams**.
- During an exam you can **ask me coding questions**. Syntax is not what is being tested.

What *is* tested: choosing the right method, running it, and explaining what the output means.

Two places to run R — the same R

RStudio, on your laptop

- What you install this week
- Where you do swirl practice and the project
- **What you use in the exams**

These slides, in the browser

- Nothing to install, opens with the deck
- What we use to practice in class
- Your typing is saved in your browser

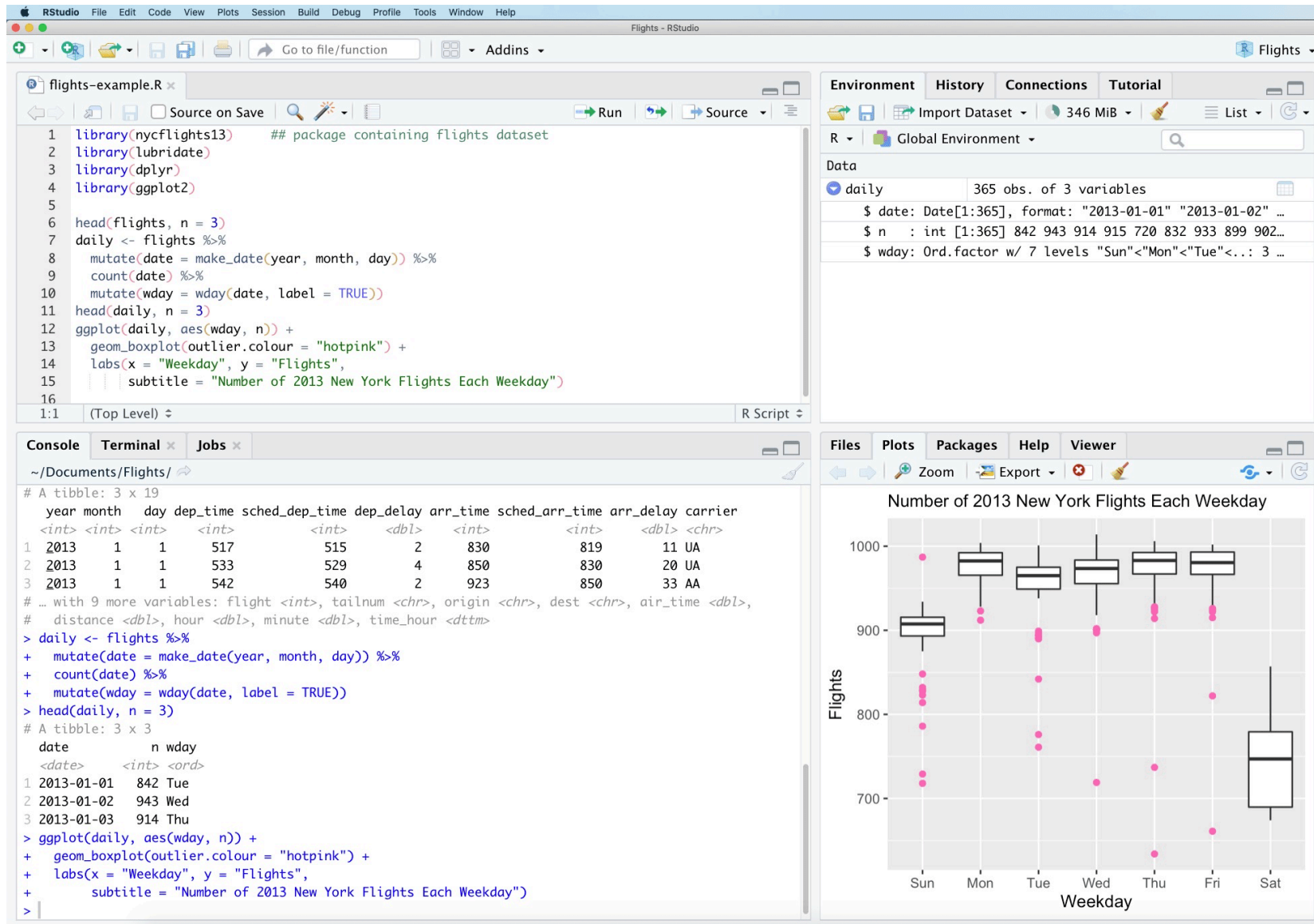
It is the **same language and the same code** — only the program around it differs. `mean(x)` is `mean(x)` in both.

Which means one thing for you

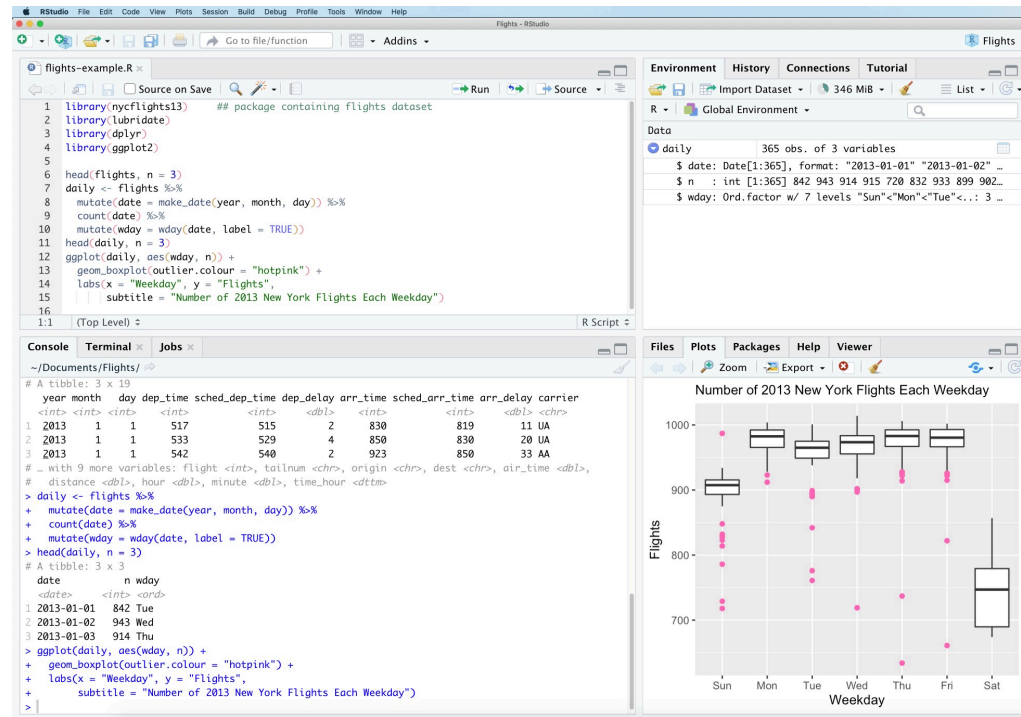
We practice **here** in class because it starts instantly and nobody gets stuck on installation.

You are examined **in RStudio**. So the R you install this week has to become familiar — that is what the **swirl** lessons are for. Type the code in both places and the difference stops mattering.

RStudio: what each pane is for



RStudio: what each pane is for



Top-left — Source / editor Where you write and save your script. Run a line with **Ctrl+Enter**.

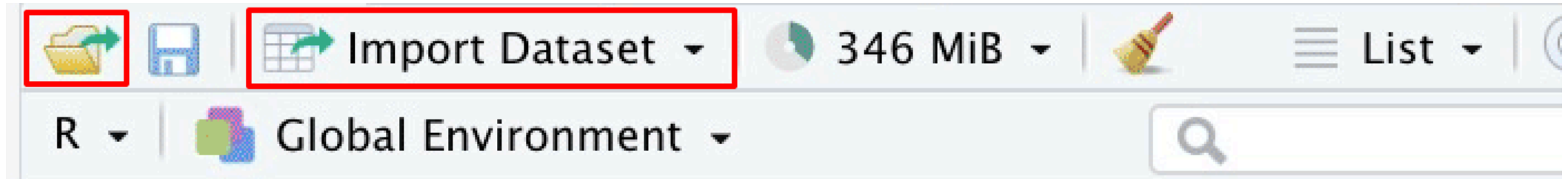
Bottom-left — Console Where code actually runs. Results and error messages print here.

Top-right — Environment Every object you have created. Your data appears here once loaded.

Bottom-right — Files · Plots · Help Your plots appear here, along with packages and help pages.

Loading data — the buttons

Both live at the top of the **Environment** pane:



- The **folder icon** opens an `.RData` / `.Rda` file — click it, pick the file, done.
- **Import Dataset** is for `.csv` and Excel files — it shows you a preview, and the code it will run, before you press Import.

Whichever you click, RStudio writes the line for you and runs it in the Console.

Loading data — the line of code

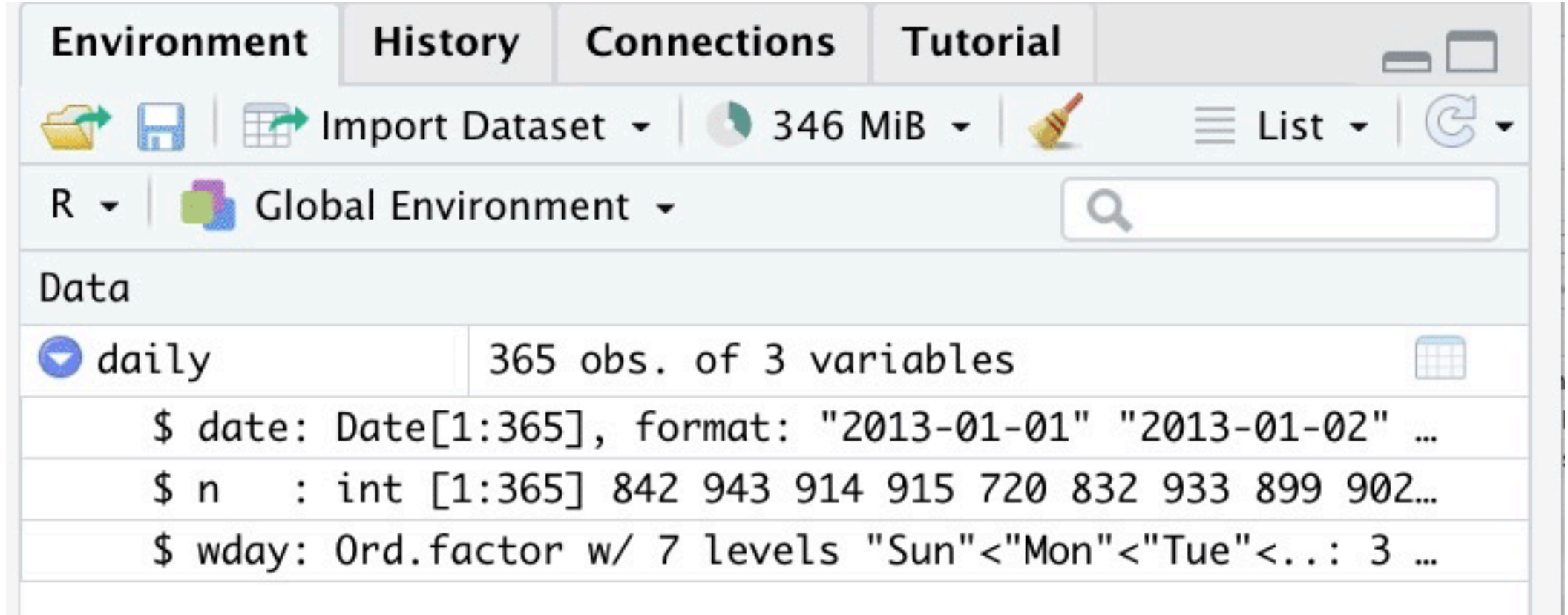
The buttons only save typing. The line itself is what you keep in your script:

```
load("Data/workforce.Rda")      # an .Rda file  
mydata <- read.csv("Data/workforce.csv")  # a .csv file
```

Put that line in your script and your analysis reruns from scratch — clicking through the menus every time does not.

Where the data ends up

Either way, look at the **Environment** pane. If the object is listed, it worked:



The screenshot shows the RStudio Environment pane. At the top, there are tabs for 'Environment', 'History', 'Connections', and 'Tutorial'. Below the tabs is a toolbar with icons for file operations and a 'List' button. The main area of the pane shows the 'Global Environment' with a search bar. Under the 'Data' section, the object 'daily' is listed with a dropdown arrow and a calendar icon. To the right of 'daily' is the text '365 obs. of 3 variables'. Below this, the structure of the object is displayed:

```
$ date: Date[1:365], format: "2013-01-01" "2013-01-02" ...  
$ n    : int [1:365] 842 943 914 915 720 832 933 899 902...  
$ wday: Ord.factor w/ 7 levels "Sun"<"Mon"<"Tue"<...: 3 ...
```

Here `daily` is loaded: **365 obs. of 3 variables**. Now you can type its name and use it.

When it doesn't work

If R answers “*cannot open file ... No such file or directory*”, the file is fine — R is **looking in the wrong folder**.

- Check where R currently is: `getwd()`
- Point it at your course folder: **Session → Set Working Directory → Choose Directory**
- Better: open the course folder as an **RStudio Project**, and the path is always right.



Exercise 0 — The same code, right here

Prove it to yourself. Run these three lines below, then run the identical lines in RStudio later today — same answers, different window.

Exercise

↺ Start Over

⚠ Show Solution

▶ Run Code

1 `2 + 2`

2

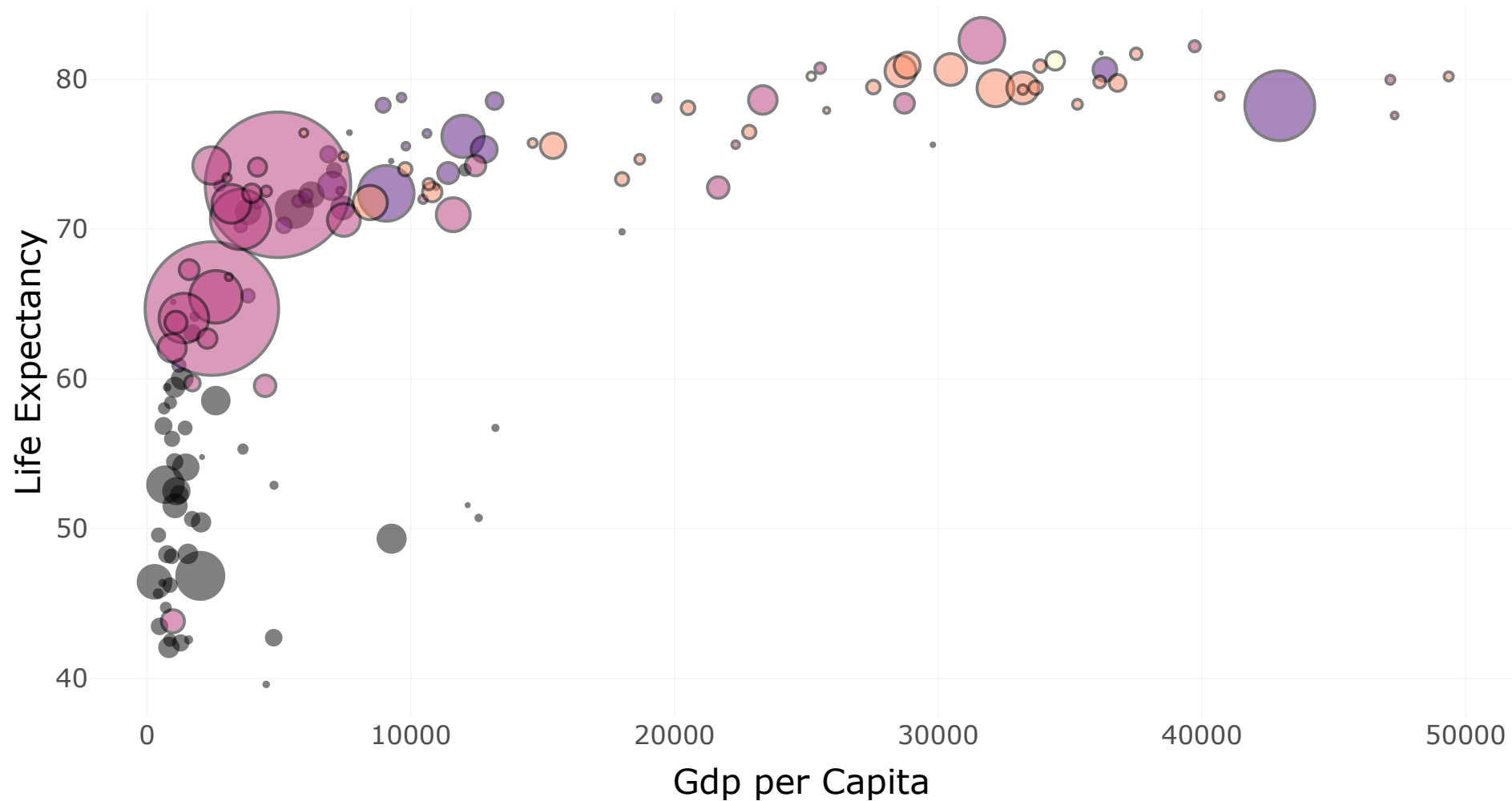
3 `x <- c(10, 20, 30, 40)`

4 `mean(x)`

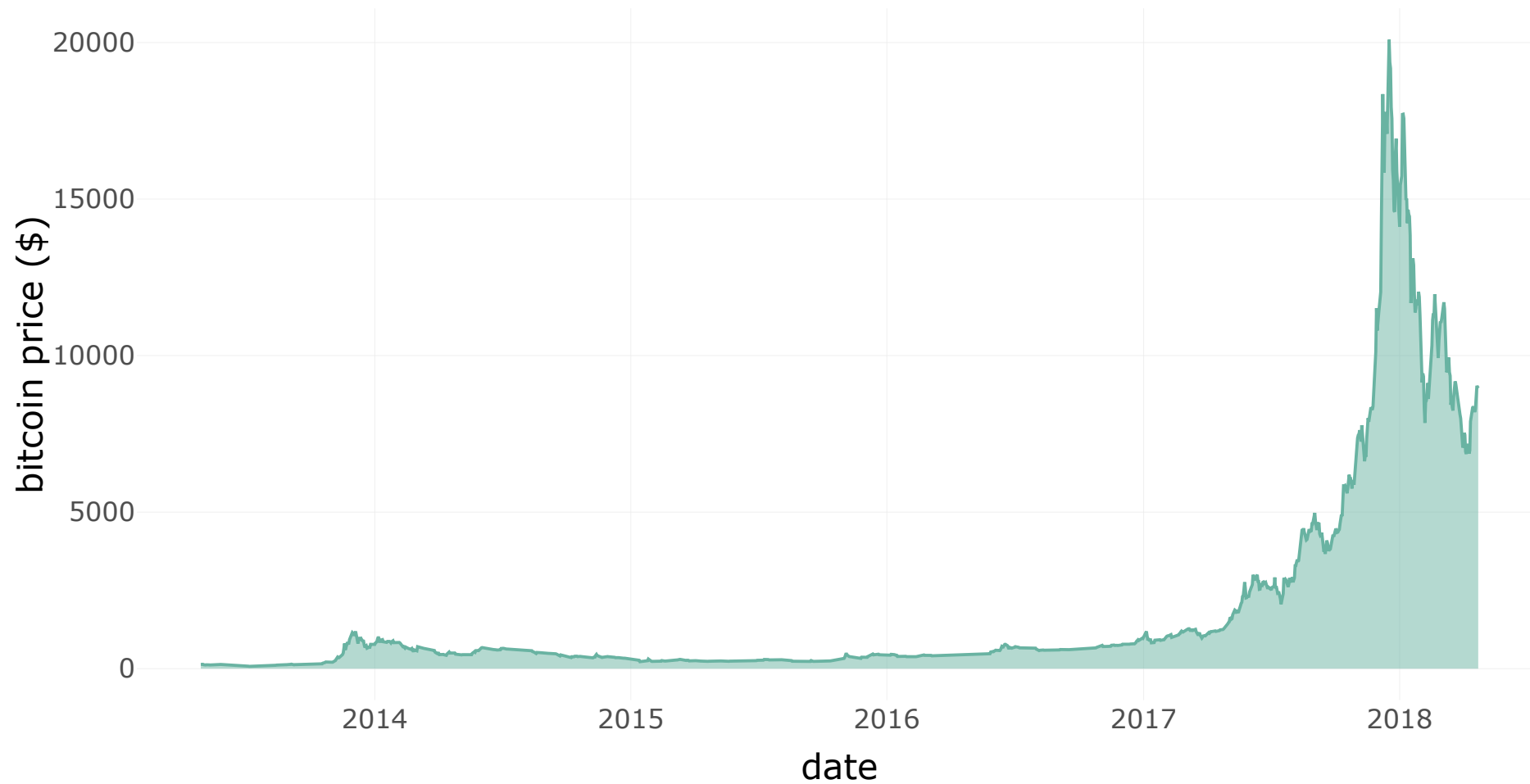
5

6 `sqrt(144)`

Life expectation vs GDP per Capita



Bitcoin price in time



Remarks

- Knowledge of theoretical statistics does not matter much if you can't apply it using modern tools
- We will use it for practical exercises with data
- You will use it for the final project
- You will use it for the applied exams
- We will learn some of it together in class

Introduction to Forecasting

Tools will often depend on the horizon and data availability

Forecasting Horizons

- **Very Short Horizon:**
 - High-Frequency Trading: Real-time price predictions for financial trading
- **Short Horizon:**
 - Public bikes: Predict the availability of bikes at bike station and adjust the number
- **Long Horizon:**
 - New obesity drug: Forecasting number of potential patients and their resources

Overview of Forecasting Techniques

1. Qualitative Forecasting

- Based on subjective judgment and expert opinions
- Suitable for unique situations or new markets
- Examples: economic impact of an oil-price change, political stability in a region

2. Quantitative Forecasting

- Uses historical data and numerical techniques
- Suitable when data is available and past patterns carry into the future
- Examples: demand for a product you already sell, next quarter's revenue

We take them in that order: **qualitative first**, then quantitative — which is where the rest of the course lives.

Methods of Qualitative Forecasting

The oracle of Delphi



Delphi Method

A structured process to reach consensus on complex, uncertain, long-term questions:

1. Select a group of experts.
2. Keep them **anonymous** — they don't talk to each other.
3. Ask them to answer a questionnaire.
4. Collect the initial responses.
5. Compile them into a summary.
6. Send the summary back; collect **refined** answers.
7. Repeat until consensus (or no further improvement).

Delphi Method — example

Determining AI threats.

- *Question:* what are the biggest risks from AI development?
- *Panel:* computer scientists, engineers, AI-company CEOs, ethicists.
- Send questionnaires on potential threats → compile → send back.
- Several rounds until the panel converges.
- *Output:* a ranked list of the most probable risks.

Brainstorming

A creative technique for generating ideas — free thinking, building on others.

- Form a group (**no need for experts**).
- State the problem clearly.
- Encourage ideas, no matter how crazy.
- Combine and build on each other's ideas.
- Document, then synthesize.

Example — employee engagement: a tech firm's HR gathers reps from HR, IT, and other teams to generate ideas for an app that boosts engagement, then implements the most promising ones.

Panel of Experts

Assemble knowledgeable people **in the same place, at the same time.**

- They meet, share insights, and discuss openly.
- Helps reach well-informed decisions.
- Often ends in a report with conclusions.

Example — environmental policy: a government agency convenes scientists, economists, conservationists, and policymakers to debate options and draft comprehensive environmental policy.

Focus Groups

Gather a **diverse** set of participants — not necessarily experts.

- Share perceptions, attitudes, and opinions.
- Produce qualitative data and consumer insight.

Example — a new TV show: before launch, a network gathers viewers from varied demographics to gauge appeal and fine-tune the show and its marketing.

The four methods — at a glance

One business use for each method:

- **Delphi** → ranking **AI threats** with an anonymous expert panel.
- **Brainstorming** → an HR team's ideas for an **employee-engagement** app.
- **Panel of Experts** → a convened panel drafting **environmental policy**.
- **Focus Groups** → viewers gauging a **new TV show** before launch.

Methods of Quantitative Forecasting

Quantitative forecasting — when can you trust it?

Quantitative methods use historical data and numerical techniques. They work when the **continuity assumption** holds:

past trends and relationships continue into the future.

When does it hold?

Interest rates and investment — the mechanism linking them is stable.

When does it break?

Covid cases and deaths, once vaccines arrived — the relationship itself changed, so the past stopped predicting the future.

Time Series Forecasting

- **Time series data**: collection of data points for a single unit (one firm, one person, one country) ordered chronologically. Can be one or more variables.
- **Time series forecasting**: identifying patterns and trends in historical data to predict future values

Year	GDP per capita (USD)
2010	9400
2011	10342
2012	10376
2013	10866
2014	11076
2015	9753
2016	8875
2017	9434

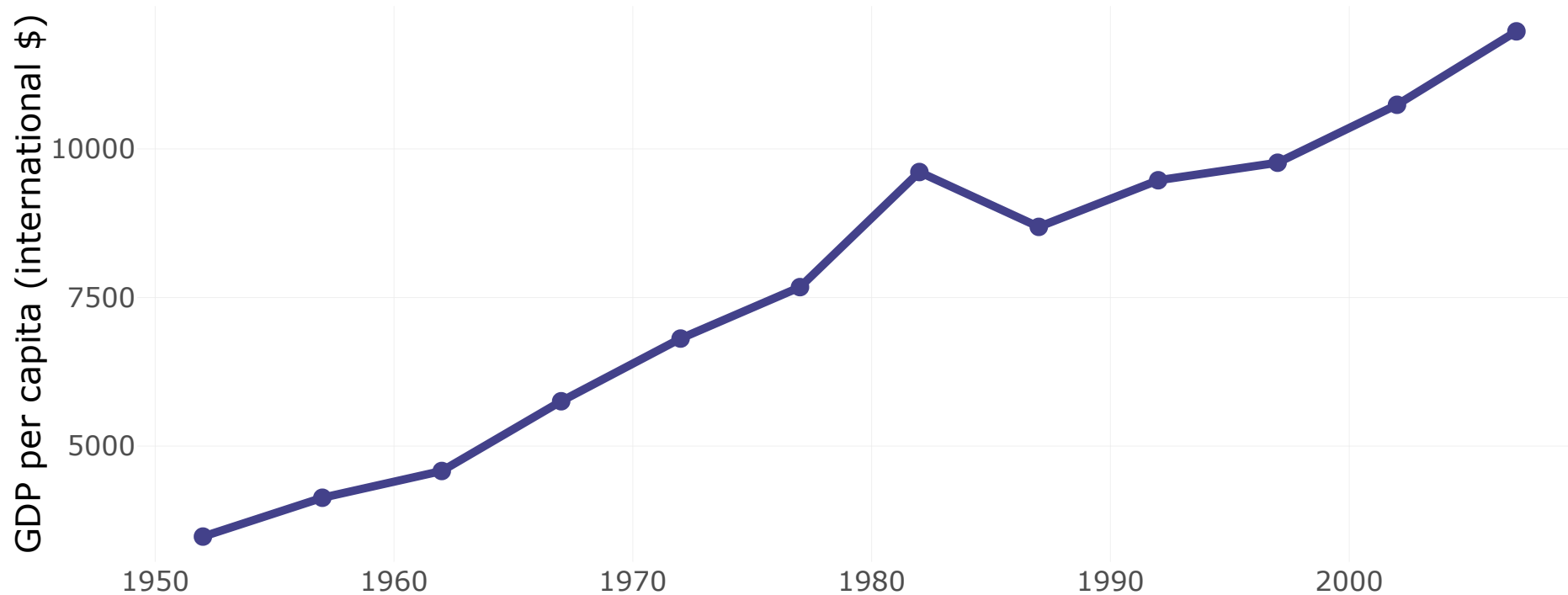
In simple terms:

- We don't care about what causes what
- We just hope that past values of the variable and its historical behavior can predict its future values

Example: Forecasting of GDP

- Time series forecasting can help predict a country's Gross Domestic Product.

Mexico: GDP per capita, 1952-2007



Hover any point. Given this history, what would you predict for the next five years?

Explanatory Models

- We have data on both the variable of interest and other variables related to it
- We consider how other variables impact the outcome of interest
- We use these relationships to make forecasts

Example: Sales at a new location

- Should we open a new Starbucks at ITAM?
- Using existing locations, analyze impact of:
 - Foot traffic
 - Neighborhood income
 - Competitors' stores
- Given these relationships, what would be sales at ITAM?

S_ID	Address	F_Traffic	N_Income	C_Stores	Sales
1	123 Main St	100	60000	2	5000
2	456 Oak Ave	150	75000	1	6000
3	789 Elm Rd	80	55000	3	4500
4	321 Maple Ln	120	80000	2	7000
5	543 Birch St	200	70000	1	8000
6	876 Pine Ave	90	60000	3	5500
7	987 Cedar Rd	180	90000	2	9000
8	654 Walnut Ln	110	85000	1	7500
9	234 Spruce St	140	75000	2	6500
10	ITAM	170	65000	1	NA

Steps of Forecasting

1. Problem Definition

- Clearly define the forecasting objective.

2. Gather Data

- Identify and collect relevant data.

3. Preliminary Explanatory Analysis

- Understand data characteristics and relationships.

4. Choosing and Fitting the Model

- Select and fit the appropriate forecasting model.

5. Evaluating the Model

- Assess the model's performance using historical data.

6. Communicate the results

- Visualize and interpret the outcomes and write a report or presentation

Remainder of the course

Quantitative Forecasting

1. Review of Statistics
2. Simple linear regression
3. Multiple linear regression
4. Time Series

Before next class — your Week 1 checklist

1. **Get R running.** Follow the **Week 1 Setup Guide** on Canvas: install R + RStudio and run the one package-install block. Bring a working laptop — from Week 2 we use R in class.
2. **Run [RStudio_practice_1.R](#)** (Canvas, ungraded). If every line runs without a red error, you are ready for Quiz 1. If one fails, bring the error message to class.
3. **Start the swirl lessons.** More practice **in RStudio** — the program you are examined in.
4. **Bookmark the [R cheatsheet](#)** — you'll use it all semester.
5. **Start thinking about a project topic** — a question you care about, with data you can get. Groups form by **1 September**.

By the end of today you should be able to: say what forecasting is, give three business decisions it informs, and explain how this course is graded.

Questions?